

## **Pablo Lara**

### **Module 2 Reflective Journal**

This project was my first time using Google Colab and working through a machine learning notebook. When I first opened Colab, it was very intimidating because I did not even know Google offered a service like this or what it was used for. There were a lot of unfamiliar tools and options on the screen, and that made me nervous because I did not understand what they did or how they could be useful.

Colab started to feel less intimidating once I began running the code. It was simple to click the play button for each cell, and having the instructions close to the code helped me understand what I was supposed to do. I realized that I did not need to understand every tool on the screen before I could start using the notebook. Working through the project one step at a time made the environment feel much more manageable.

I had also never used pandas, NumPy, or matplotlib before this project. It is still difficult for me to remember exactly what each library does, but I became interested in how they all worked together. One tool helped organize the data, another performed calculations, and another turned the data into visualizations. Seeing them used together helped me understand that machine learning projects use different tools for different parts of the process.

The visualization section was another important part of the lab for me. When I first saw the scatter plot, I felt like I had no idea what I was looking at or what the points meant. As I continued through the lab and reached the bar graph, the data started to make more sense. I could clearly see the three flower species and that each one had 50 samples. This helped me understand how graphs can make a large amount of data easier to understand than looking only at rows of numbers.

This project also changed the way I think about machine learning. Before this lab, I did not fully understand how much work could be done with data before even building a model. We were able to organize the data, calculate statistics, create graphs, compare different groups, and look for patterns. That made me curious about what else can be discovered, predicted, or understood by using different datasets and machine learning tools.

Another benefit of this project is that I am currently taking a Python class. Seeing Python used in a real machine learning project made me feel more confident in that class. Instead of only seeing

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Pablo Lara

Brandy Gonzales

Marlon Roberts

Derrick K. Gibbs-McGlaston

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small practice problems, I was able to see how Python can actually be used to work with data, perform calculations, and create visualizations. That made the code feel more meaningful and gave me a better idea of how Python can be used outside of basic exercises.

Overall, I feel much more comfortable using Google Colab now than I did at the beginning of the project. I still have a lot to learn about pandas, NumPy, matplotlib, and machine learning, but the tools no longer feel as intimidating. This lab helped me see how coding, data, statistics, and visualization can all work together, and it made me more interested in learning what else can be done with data.

## **Brandy Gonzales**

Reflective Journal- Brandy Gonzales  
Lab: Module 02- pandas, NumPy, and matplotlib (ITAI 1371)

At first this lab felt like a pile of tools with fancy names. I knew pandas, NumPy, and matplotlib were “what you use for ML,” but I didn’t really feel what each one was for until I had to use them on the same dataset in one notebook.

The biggest shift for me was pandas. Pulling a single column like sepal length and handing it straight to NumPy made something click: the table isn’t just something to stare at-it’s something you can slice and measure. When I calculated the mean and standard deviation, it stopped being abstract math and became “what’s typical for this flower measurement, and how much does it usually vary?” That same idea shows up in cybersecurity analytics too, which helped the lab feel less random and more connected to why I’m taking these classes.

The scatter plot was the moment the data became real. Reading that sets, versicolor, and Virginia were three classes was one thing. Seeing sets sit off by itself while the other two blend together was another. That visual made “easy vs hard to classify” make sense in a way a paragraph never did. I also noticed I care more about plots now, not as a decoration, but as a check: do the groups even separate before I trust a model?

I got stuck less on big concepts and more on small workflow things- running cells in order, make sure outputs showed up, and remembering that Jupyter only displays the last line unless I print. Those tiny fractions taught me as much as the content. Same with the team GitHub piece: having the notebook in a shared repo is useful, but I still have to run it myself and put my own words on the reflection, or the learning (and the credit) isn’t really mine.

What I want to carry into Module 3 is simple: load the data, look at shape and columns, measure a couple features, plot something, then ask what the picture is really telling me. I’m less intimidated by the tooling now. I’m more curious about what the patterns mean.

## Derrick K. Gibbs-McGlaston

### Tools Used in Machine Learning

Before completing this lab, I understood that Python was used in machine learning, but I did not fully understand how the different Python tools worked together. I had some familiarity with programming and data, but this lab helped me see that machine learning is not just about creating a model. A large part of the process involves preparing, examining, understanding, and visualizing data before a model is ever built. Working with the Iris dataset gave me a simple way to see how these tools fit into the larger machine learning workflow.

One useful concept for me was working with pandas DataFrames. I found the DataFrame concept easy to relate to because it is similar to working with a spreadsheet, but it provides much more flexibility for programming and analyzing information. Instead of manually examining rows and columns, I could use Python to select information, calculate statistics, group records, and inspect the structure of the dataset. The `groupby()` allowed me to calculate the average measurements for each species. This helped me understand how programming can make data analysis repeatable instead of having to do calculations manually.

The Iris dataset also helped me understand the importance of knowing what the data actually contains. The dataset had 150 samples, four numerical features, and three different classes. Seeing this information in the DataFrame made the dataset much less abstract. I could see the individual measurements and understand that each row represented an observation while the columns represented different characteristics of the flowers. This proves that machine learning models depend on the quality and structure of the data given.

Working with NumPy, calculating the mean and standard deviation of the sepal lengths showed me how calculations can be performed efficiently on collections of data. The mean sepal length was 5.84 centimeters, while the standard deviation was 0.83 centimeters. Rather than calculating these values manually, NumPy allowed the calculation to be performed directly on the data. This helped me understand why numerical libraries are so important when working with larger datasets.

The visualization portion of the lab was probably the part that made the relationship between data analysis and machine learning make sense. When the sepal length and sepal width were displayed in a scatter plot, I could visually compare the three species. The setosa samples appeared more separated from the other species, while versicolor and virginica showed more similarities. Looking at the graph made patterns easier to recognize than they would have been by looking only at a table of numbers. This showed me that visualization is not simply about making a project look better. It can be an important analytical tool for discovering relationships and patterns in data.

Another lesson was the importance of checking my work instead of assuming that code is correct because it runs. The notebook included verification steps, such as checking that the mean and standard deviation were numerical values. Running the entire notebook from beginning to end also helped demonstrate the importance of reproducibility. A machine learning project should be able to run consistently rather than depending on calculations or steps that are performed manually.

The lab also helped me understand how these tools connect to future machine learning work. Pandas can be used to organize and clean datasets, NumPy can perform numerical operations, matplotlib can help visualize patterns, and scikit-learn provides tools for building machine learning models. Understanding that relationship makes the next stages of the course easier to understand because I can see where data preparation and exploration fit before a model is trained.

One question I am interested in exploring further is which features provide the most useful information for distinguishing between different classes. The Iris dataset contains both sepal and petal measurements, and I would like to investigate whether some measurements are much more useful than others. I am also interested in seeing how the results of a visualization compare with the actual performance of a machine learning model.

This lab changed my understanding of machine learning tools from seeing them as separate Python libraries to seeing them as components of one workflow. I learned that understanding the data comes before trying to make predictions from it. Good machine learning begins with disciplined data analysis: knowing what the data contains, checking its quality, finding meaningful patterns, and using the right tools to understand it before moving on to modeling.

## **Marlon Roberts**

My role was learning the basic things about the 16-cell notebook, learning how to read the graph and how to execute it. Coming into brand new it looked very confusing to me, once I slowly broke it down and knew what I was looking at and looking for it was all slowly making sense. Once you get the basic information on what the tools do you will get an understanding with it. I analyze the data set and extract the data that is needed. I now know there are 3 things to look for in a data set: class, features and samples. NumPy means numerical python. Cleaning data must be very precise because one wrong number can mess you up. I will be getting used to seeing a lot of data frames and knowing what data sets are. I think the next time I attempt this I will be a lot better and know what I'm looking for in the data frame.